

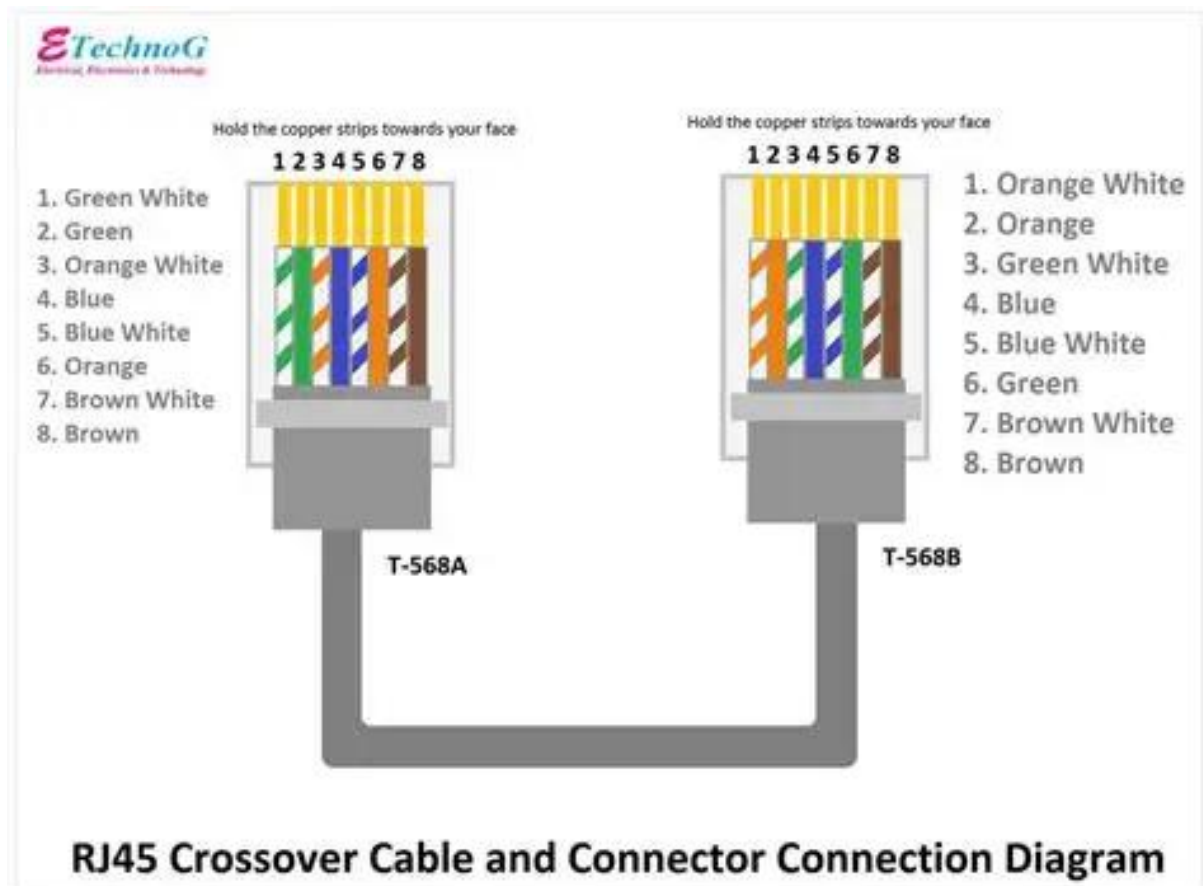
TASK: 1

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	Ethernet Cable	Fiber Optic Cable
1. Transmission	Uses electrical signals to transfer data.	Uses light signals to transfer data.
2. Speed & Distance	Suitable for shorter distances and common network connections.	Supports very high speeds and much longer distances.
3. Cost	Generally cheaper and easier to install.	Generally more expensive and requires specialized tools for installation.

TASK: 2

Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



TASK: 3

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Steps to Attach an RJ45 Connector to a CAT6 Cable

1. Cut the CAT6 cable to the required length and remove about 2–3 cm of the outer jacket.
2. Separate and straighten the eight individual wires inside the cable.
3. Arrange the wires according to the **T568B standard**: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
4. Straighten the wires carefully and trim their ends evenly.
5. Insert all eight wires into the RJ45 connector, making sure each wire reaches the correct pin and the cable jacket goes slightly inside the connector.
6. Check the connector from the front to make sure the wires are in the correct order.
7. Place the RJ45 connector into the **RJ45 crimping tool**.
8. Squeeze the crimper firmly to secure the connector and lock the wires in place.
9. Repeat the same process on the other end if a complete Ethernet cable is required.
10. Finally, test the cable using a **network cable tester** to confirm that all eight connections are working correctly.

TASK: 4

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

For a gaming café with 10 PCs, I would choose **Ethernet cables (CAT6)** because they provide good speed and low latency for LAN gaming at an affordable cost. The distances between the PCs and network switch are usually short, so fiber optic cables are not necessary. I would use an **RJ45 crimper** to attach RJ45 connectors to the Ethernet cables because it is simple, inexpensive, and suitable for setting up multiple PCs. This setup would provide reliable performance while keeping the overall installation cost low.